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formed of sensory hair cells and supporting cells. The function of these canals is to detect rotational movement of the head.

10.6.3 Olfactory receptors

The receptors for taste and smell are the chemoreceptors. The smell receptors are excited by air borne chemicals that dissolve in fluids. The yellow coloured patches of olfactory epithelium form the olfactory organs (figure.10.22) that are located on the roof of the nasal cavity. The olfactory epithelium is covered by a thin coat of mucus layer below and olfactory glands bounded connective tissues, above. It contains three types of cells**: supporting cells, Basal cells** and millions of pin shaped **olfactory receptor cells** (which are unusual bipolar cells). The olfactory glands and the supporting cells secrete the mucus. The unmyelinated axons of the olfactory receptor cells are gathered to form the filaments of olfactory nerve [cranial nerve I] which synapse with cells of olfactory bulb. The impulse, through the olfactory nerves, is transmitted to the frontal lobe of the brain for identification of smell and the limbic system for the emotional responses to odour.

Gustatory receptor: The sense of taste is considered to be the most pleasurable of all senses. The tongue is provided with many small projections called **papillae** which give the tongue an abrasive feel. Taste buds are located mainly on the papillae which are scattered over the entire tongue surface.

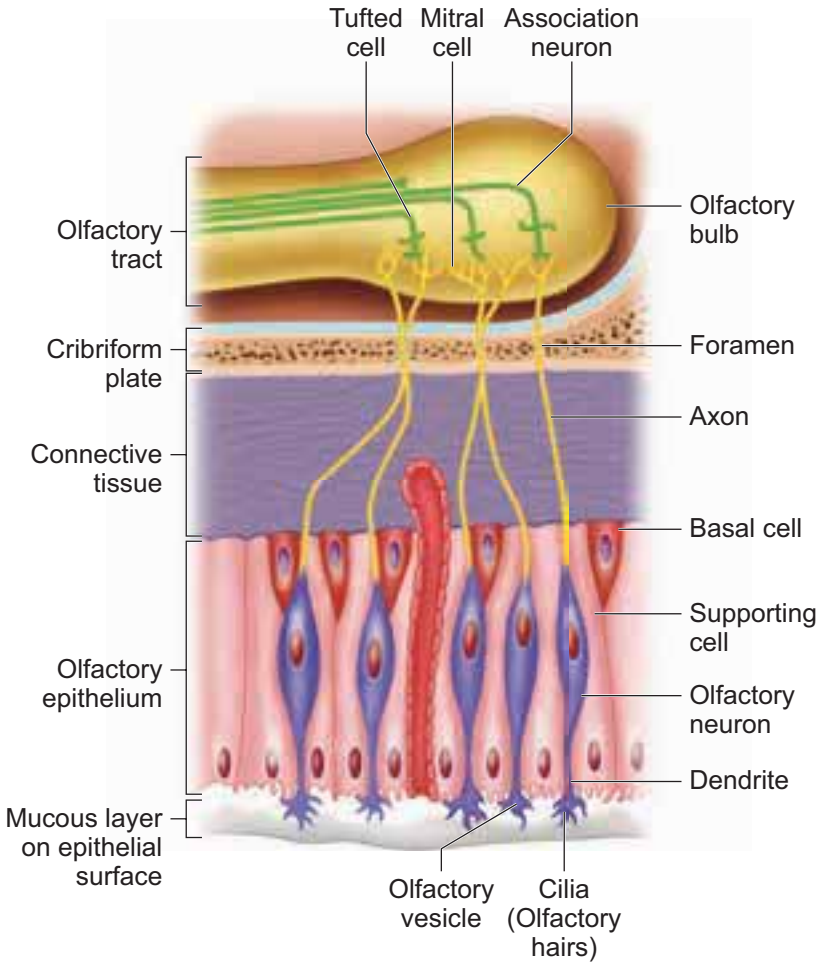

Figure 10.22 Olfactory organ